

11/17 Group Meeting Twisted sheaves and Twisted FM. □

Ref ① A. Căldăraru (11/15/12) = first systematic treatment
 "Derived categories of twisted sheaves on Calabi-Yau manifolds" (00)

"non-Fine Moduli spaces of sheaves on K3 surfaces" (02)

② D. Huybrechts & P. Stellari = twisted Chern char.
 "Equivalences of twisted K3 surfaces" (05)

③ K. Yoshioka = "moduli"

"Moduli spaces of twisted sheaves on projective variety" (06)

④ M. Lieblich (11-7/13) = "Stack theoretic"

"Moduli of twisted sheaves" (07)

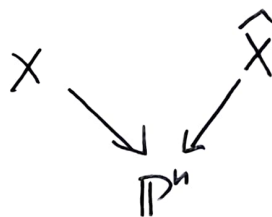
Rough Motivation

X : IHS w/ Lag fib

$\hat{X}$: dual fib (if exists)

$$\Rightarrow D^b(X) \cong D^b(\hat{X})$$

Bridgeland - Maciocia	for ellip fib. for K3, Abel fib.
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$\hat{X}$: usually const
 via moduli of
 sheaves on X
 $\Downarrow$

moduli prob.

* In order to treat the case of

- fibration with no section
- Non-fine moduli

$\Rightarrow$ Căldăraru, Sawon... use Twisted univ sheaf

$$\& \mathbb{P}: D^b(\hat{X}, \alpha) \xrightarrow{\sim} D^b(X)$$

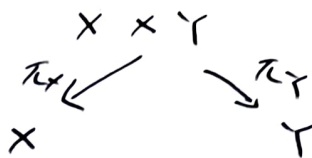
Untwisted Case?

[2]

Integral functors (h, Mukai)

X, Y : sm proj

$P \in D^b(X \times Y)$: obj



$$\begin{aligned} \Phi_{Y \rightarrow X}^P : D^b(Y) &\rightarrow D^b(X) \\ \varepsilon \mapsto R\pi_{X*} (P \otimes^L \pi_Y^* \varepsilon) \end{aligned}$$

← derived tensor prod.

↑ pull back

↓ derived pushforward along π_X

If it is an equiv of cat.
 $\xLeftrightarrow{\text{def}}$ FM transform.

Examples

Mukai A : Abelian var. $\hat{A} := \text{Pic}^0 A$

$\Rightarrow \exists P$: Poincaré line bundle / $A \times \hat{A}$

$$\Phi_{\hat{A} \rightarrow A}^P : D^b(\hat{A}) \xrightarrow{\sim} D^b(A)$$

$\{X(\mathcal{O}(m))\}$

more common

Mukai S : K3. $v \in \hat{H}(S, \mathbb{Z}) = H^0 \oplus H^2 \oplus H^4$: w -primitive.

w : ample $v^2 = 0$

$\Rightarrow M := M_w^{S(S)}(v)$: K3 again $w/\mathcal{U}$: univ sheaf.

$$\Phi^{\mathcal{U}} : D^b(M) \xrightarrow{\sim} D^b(S)$$

Facts

Orlov X, Y sm proj. $\Phi : D^b(X) \rightarrow D^b(Y)$ equiv (or fully faithful)
 $\Rightarrow \exists ! P \in D^b(X \times Y)$ up to isom $\Phi \cong \Phi^P$

eg. shift $\Phi^{\mathcal{O}_\Delta[n]} \cong [n]$

pull back $\Phi^{\mathcal{O}_{T(f)}} \cong f^*$

line bundle $\Phi^{\mathcal{L}(-)} \cong \mathcal{L} \otimes (-)$

$\mathcal{L} : X \rightarrow \Delta \subset X \times X$

① Passage to cohomology

- $H^*(X, \mathbb{Q}) = \bigoplus H^i(X, \mathbb{Q})$ total cohomology

- $D^b(X) \xrightarrow{[-1]} K(X) \xrightarrow{\text{Grothendieck gr}} H^*(X, \mathbb{Q})$ Mukai vector

$\varepsilon \mapsto \text{ch}(\varepsilon) \cdot \sqrt{\text{td}_X}$ cup prod.
 $\sum (-1)^i \text{ch}(\varepsilon^i)$

- $D^b(X) \xrightarrow{\bar{\Phi}_P} D^b(Y)$ underlying idea
 $\downarrow \cup \quad \downarrow \cup$
 $H^*(X, \mathbb{Q}) \xrightarrow{\bar{\Phi}_P^H} H^*(Y, \mathbb{Q})$ $e \mapsto \pi_{X*}(\cup(P) \cdot \pi_Y^* e)$
 } - additive
 } $\text{ch}(L) = \exp(C_1(L))$

- Mukai for K3, the induced map defines a Hodge isometry

$\tilde{H}(X, \mathbb{Z}) \xrightarrow{\sim} \tilde{H}(Y, \mathbb{Z})$

$\bar{\Phi}_\varepsilon^H$ will not preserve the cohomological degree

$\Rightarrow$ one should not expect $H^2(X, \mathbb{Z}) \cong_{\text{Hodge}} H^2(Y, \mathbb{Z})$

Orlov for Abelian var

$(\Rightarrow X \cong Y)$

$\leadsto$ Torelli type Thm

② Criterion for equiv

Bondal - Orlov - Bridgeland - Mukai

X, Y : sm proj $\bar{\Phi}^P: D^b(Y) \rightarrow D^b(X)$ Int. funct.
 $\bar{\Phi}^P$ equiv iff $(P_y = \bar{\Phi}^P(\mathcal{O}_x))$

i) $\text{End}_X(P_y) = \mathbb{C}$

ii) $\text{Ext}_X^i(P_{y_1}, P_{y_2}) = 0$ $\forall y_1 \neq y_2$ or $i \notin [0, \dim X]$

iii) $P_y \otimes \omega_X \cong P_y$ preserves Serre
 $(\bar{\Phi}^P \cdot S_Y \cong S_X \cdot \bar{\Phi}^P)$

} strongly simple
 $\Downarrow$
 fully faithful

$\{\mathcal{O}_y\}$: spanning class $\leadsto$ orthogonal basis w.r.t Ext pairing

Twisted? so far we have considered only fine moduli [14]
2-obstruction of global univ sheaf $\boxed{\mathcal{E}|_{\mathbb{A}^1 \times X} \cong \hat{\mathcal{X}}}$

① univ sheaf may not exist locally

If $\exists$ strict semistable $\Rightarrow$ represents more than 2 sheaves
(all S -equiv classes)

$\Rightarrow$ We will assume stable = semistable
by choosing nice polarization.

② Although $\exists$ univ sheaf locally

They may not glue well along all of M

$\rightarrow$ We need to twist the gluing cocycle
(modify)

$\Rightarrow$ Twisted Univ sheaf?

first introduced by Giraud ($\exists$ D-) 1971
as part of study on non-comm homology.

Def $(X, \mathcal{O}_X)$: scheme (w/ étale top)

or analytic space (w/ étale or analytic top)

Let $\alpha \in \check{C}^2(X, \mathcal{O}_X^*)$ Čech 2-cocycle.

Let $U = \{U_i\}$ open cover. $\alpha_{ijk} \in \Gamma(U_i \cap U_j \cap U_k, \mathcal{O}_X^*)$ section

α -twisted sheaf on X $(\{\mathcal{F}_i\}, \{\varphi_{ij}\})$

$\mathcal{F}_i$: $(\text{coh})^{\text{Coh}}$ sheaf of $\mathcal{O}_X$ -module on U_i

$\varphi_{ij}: \mathcal{F}_i|_{U_{ij}} \rightarrow \mathcal{F}_j|_{U_{ij}}$ isom

s.t 1) $\varphi_{ii} = \text{id}$ 2) $\varphi_{ij} = \varphi_{ji}^{-1}$

3) $\varphi_{ij} \cdot \varphi_{jk} \cdot \varphi_{ki} = \alpha_{ijk} \cdot \text{Id}$.

Up to equiv cat. independent of choice open cover, cocycle.
& form an abelian cat. $\text{Coh}(X, \alpha) \xrightarrow{\sim} \mathcal{D}^b(X, \alpha)$.

Brauer group = group of projective-linear twisting.

[5]

cohomological $Br(X) = H^2_{\text{ét}}(X, \mathcal{O}_X^*)$ or $H^2_{\text{ét}}(X, G_m)$

Idea (c.f. Severi-Brauer var, G_m -gerbe) use Kummer

$M^S = \text{Quot}^S // PGL_n \quad \exists \mathcal{E} \text{ on } \text{Quot}^S \times X$

Q Can we descend $\mathcal{E}$ to $M^S \times X$?

$0 \rightarrow \mathcal{O}_M^* \rightarrow GL_n \rightarrow PGL_n \rightarrow 0$

$\rightarrow H^1(M, GL_n) \rightarrow H^1(M, PGL_n) \xrightarrow{\delta} H^2(M, \mathcal{O}_M^*) \rightarrow 0$
coboundary

geometric $Br(X) = \text{image} \left(\bigcup_{n \geq 1} H^1_{\text{ét}}(X, PGL_n) \xrightarrow{\delta} Br'(X) \right)$

- X : sm proj (Gabber) $\Rightarrow Br(X) = Br'(X)_{\text{tors}}$
or sm surf (Gruenhoeck, de Jong)

- $\exists \alpha$ -twisted loc tree $\Leftrightarrow \alpha \in Br(X)$

- $Br(X) = \{ \text{Azumaya alg} \} / \text{Morita equiv.}$ $A \otimes \text{End}(E) \sim A$
 $\hookrightarrow \mathcal{O}_X$ -alg coh as $\mathcal{O}_X$ -mod. E : loc tree
étale locally $\cong M_n(\mathcal{O}_X)$

α -twisted loc tree $\mathcal{F}_1$

$A := \text{End}(\mathcal{F}_1)$ (usual) coh sheaf ($\because$ cancel twist)
acycle.

A : non-comm alg $n = \text{ind}(\alpha)$.

We have $\text{Coh}(X, \alpha) \cong \text{Coh}(X, A)$ ($\leadsto$ period-index problem)

(analytic)
 $0 \rightarrow H^2(X, \mathbb{Z}) / NS(X) \otimes \mathbb{Q}/\mathbb{Z} \rightarrow Br(X) \rightarrow H^3(X, \mathbb{Z})_{\text{tor}} \rightarrow 0$
transcendental lattice $H_1(X, \mathbb{Z})_{\text{tor}}$ Poincaré dual

k^3 $(\mathbb{Q}/\mathbb{Z})^{22-p}$ F_{irr} $\mathbb{Z}/2$ $\text{rat. ruled } 0$
 A_{hel} $(\mathbb{Q}/\mathbb{Z})^{6-p}$ biellip $\mathbb{Z}/2, 3, 4, 6 \dots$ $\odot$ birat. inv.

Twisted Integral trans & FM M : not fine. [6]

$\mathcal{U}$: α -twisted univ $\alpha \in \text{Pr}(M)$

$\bar{\alpha} = \pi_M^* \alpha$ $\mathcal{U} \in D^b(X \times Y, \bar{\alpha}^{-1})$

$\Rightarrow \Phi_{M \rightarrow X}^{\mathcal{U}}: D^b(M, \alpha) \rightarrow D^b(X)$

④ Căldăraru $\mathcal{U}$: not necessarily w -prim., $\mathcal{U}^2 = 0$

Suppose $M^S = M^{SS}$

$\Rightarrow \Phi_{M \rightarrow X}^{\mathcal{U}}: D^b(M, \alpha) \xrightarrow{\sim} D^b(X)$

Orlov type thm

by Canonaco - Stellari (07)

BOBM type criterion

by Căldăraru

① Passage to cohomology

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 0$$

$$H^2(X, \mathcal{O}_X) \rightarrow H^2(X, \mathcal{O}_X^*) \rightarrow H^3(X, \mathbb{Z})$$

If $H^3(X, \mathbb{Z}) = 0$ (e.g. $K3$) $\Rightarrow B \in H^2(X, \mathbb{Q})$ rat B-field

$B^{0,2} \in H^2(X, \mathcal{O}_X)$ maps to α $\exp(B^{(0,2)}) = \alpha$

$\Rightarrow \text{ch}^B = e^B \cdot \text{ch}: K(X, \alpha) \rightarrow H^*(X, \mathbb{Q})$

(Note $B \in (1,1) \cap \mathbb{Z} \Rightarrow B = c_1(L)$ non-twist.)

For $K3$ (w/ by Căldăraru, by Huybrechts & Stellari)

$\Phi: D^b(X, \alpha) \xrightarrow{\sim} D^b(X', \alpha')$

$\Rightarrow \Phi_*: \tilde{H}^B(X, \mathbb{Z}) \xrightarrow{\sim} \tilde{H}^{B'}(X', \mathbb{Z})$

For Abelian surf

Zhiyuan-Li-Haitao Zou (2021) $\leadsto$ higher dim open.